



6.8.5 Wrap-Up: Ticket Out the Door

1. Evaluate each expression.

a. $447 - (-702)$

b. $-834 - (-610)$

c. $-309 - 525$

Unit 6, Lesson 9: Multiplying Integers Using Models



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



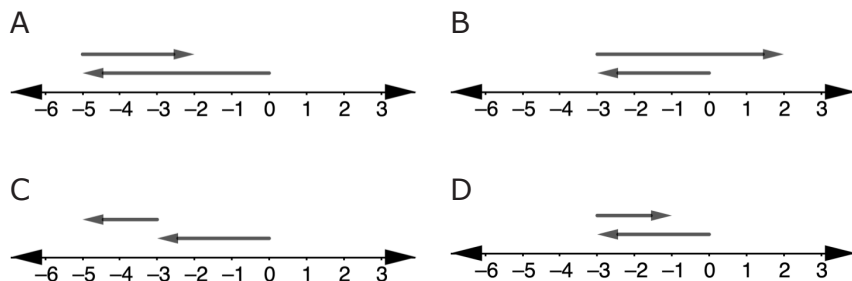
Learning Target

- ☐ I can multiply integers using integer chips, number lines, and patterns.



6.9.1 Warm-Up: Which One Doesn't Belong?

1. Four number lines modeling addition are shown.



Explain which one doesn't belong.



6.9.2 Exploration: Using Integer Chips to Represent Multiplication

1. The addition expression $7 + 7 + 7$ can be written as ____ $\cdot 7$.

This is because repeated addition of integers can be

expressed as

- ☐ subtraction.
- ☐ multiplication.
- ☐ division.

2. The expression 5×2 can be interpreted as adding ____ groups of ____ or adding ____ groups of ____.

3. Use integer chips to represent one of these interpretations.

Recall that $\ominus = -1$ and $\oplus = +1$.

Draw a picture of your representation.

4. Describe how the integer chips represent the value of the expression 5×2 .

5. The expression $5 \times (-2)$ can be interpreted as adding _____ groups of _____ or adding _____ groups of _____.
6. Explain which interpretation of $5 \times (-2)$ you think would be easier to represent using integer chips.
7. Discuss your choice with your partner. Then, work together to use integer chips to represent one of the interpretations. Draw a picture of the representation.
8. Describe how the integer chips represent the value of the expression $5 \times (-2)$.
9. What is the value of the expression $5 \times (-2)$?



Commutative Property of Multiplication

$$a \times b = b \times a$$

The **commutative property of multiplication** states that the order in which multiplication is performed does not change the value of the expression.

10. Using the commutative property of multiplication, what is the value of the expression -2×5 ?



6.9.3 Exploration: Multiplying a Negative Integer by a Negative Integer

1. Complete the statement.

The expression, $(-5)(-2)$, indicates that _____ groups of _____ should be **removed**.

2. Use the steps below to discover how to represent $(-5)(-2)$ using integer chips.

Step 1: First, use your integer chips to create the set of zero pairs that will be needed to find $(-5)(-2)$. Draw the groups of zero pairs in the box below.



Step 2: Remove 5 groups of 2 negative integer chips.

Step 3: Remove any remaining zero pairs, if needed.

3. Complete the statement.

The value of expression $(-5)(-2)$ is _____.

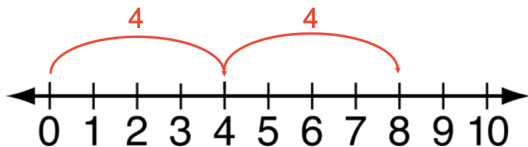
4. With your partner, discuss whether or not you think the product when multiplying two negative values will always be positive. Explain your thinking.



6.9.4 Guided Instruction: Multiplying Integers Using the Number Line

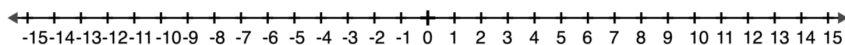
Recall that $(2)(4)$ means 2 groups with 4 units in each group. The product is the total number of objects.

This can be represented on the number line as shown.



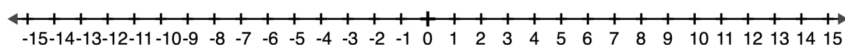
1. Represent each multiplication expression on the number line. Then, use it to complete the equation.

$$(2)(-4)$$



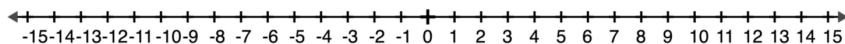
$$(2)(-4) = \underline{\hspace{2cm}}$$

$$(-2)(4)$$



$$(-2)(4) = \underline{\hspace{2cm}}$$

$$(-2)(-4)$$



$$(-2)(-4) = \underline{\hspace{2cm}}$$



6.9.5 Guided Practice: Multiplying Integers Using Models

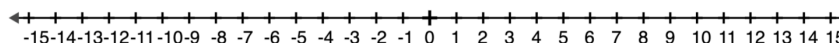
1. Use integer chips to model and evaluate each expression.

- a. $(3)(-5) =$

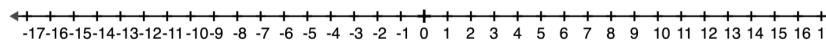
- b. $-2 \times -2 =$

2. Use a number line model to evaluate each expression.

- a. $(8)(-1) =$



- b. $(-4)(-4) =$



3. Which model for multiplication of integers do you prefer: multiplication chips or the number line?



6.9.6 Your Turn!

1. Find the value of each expression using your preferred method.

a. 2×-8

b. $-6 \cdot (-3)$

c. $(-7)(-6)$

d. $5(-3)$

e. -9×-4

f. $7 \cdot 5$



6.9.7 Wrap-Up: Cool Down

1. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
 $\times$ 	
 $\times$ 	
 $\times$ 	
 $\times$ 	